INCH-POUND
MIL-PRF-62494C(AT)
26 October 1995
SUPERSEDING
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PERFORMANCE SPECIFICATION

SHOCK ABSORBER, DIRECT ACTING

This specification is approved for use by U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance, test, and manufacture requirements for the direct acting shock absorber for use on military tracked vehicles in the Bradley Fighting Vehicle (BFV) Family.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other section of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

DEPARTMENT OF DEFENSE

12369308 - Shock Absorber, Direct Acting.
12350824 - Paint System, Chemical Agent Resistant Coating.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents that are adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the non-government documents that are current on the date of the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D2000 - Rubber Products in Automotive Applications (DoD Adopted).
ASTM D3399 - Standard Specification for General-Purpose Synthetic Detergent Liquid.

(Application for copies may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshochocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a first article sample shall be subjected to first article inspection in accordance with 4.2.

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3.2 Materials. Materials used shall be in accordance with the manufacturer's materials specifications for shock absorbers. The materials shall be capable of meeting all the operational and environmental requirements specified herein. Recovered materials shall be used to the maximum extent practicable (see 4.4.1).

3.2.1 Dissimilar metals. The shock absorber shall be designed and fabricated to minimize galvanic corrosion in dissimilar metal couples (see 4.4.1).

3.2.2 Bearing. The self-aligning ball bearing shall be fabricated from materials that are inherently corrosion resistant or are permanently treated to prevent forms of corrosion and deterioration which affect the bearing's performance. The balls shall be hardened to Rockwell C40 minimum (see 4.4.1).

3.2.3 Bearing seats. The internal bearing seats between the ball and the outer bearing shell shall be self-aligning, permanently lubricated, and be of a resilient nature to prevent brinelling (see 4.4.1).

3.2.4 Rubber components. Rubber components under stress in usage and/or subject to ozone attack which causes impairment of function and durability, shall be ozone resistant and shall conform to ASTM D2000, C12 requirements. Rubber components shall be resistant to petroleum oils and lubricants, servicing fluids and cleaning fluids where their intended use requires it (see 4.4.1.1).

3.2.5 Thermal design. All components shall be of material and design sufficient to withstand intermittent operation at an oil temperature of 580 degrees Fahrenheit (°F) (305 degrees Celsius (°C)) (see 4.4.1).

3.2.6 Hardened valve components. All contacting or sliding valve surfaces shall be either through hardened to Rockwell C50 minimum, or surface hardened to a minimum surface hardness of 69 on the Rockwell 30N scale (see 4.4.1).

3.2.7 Reserve tube durability. The reserve tube shall have a minimum yield of 60 psi (414 MegaPascal (MPa)) and wall thickness of 0.129 to 0.139 inch (3.28 to 3.53 millimeters (mm)) to prevent denting (see 4.4.1).

3.2.8 Dirt shield tube durability. The dirt shield tube shall have a minimum yield of 60 psi (414 MPa) and wall thickness of 0.076 to 0.088 inch (1.93 to 2.24 mm) to prevent denting (see 4.4.1).

3.2.9 Piston rod hardness. The piston rod surface hardness shall be Rockwell C35 through C47 (see 4.4.1).

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3.3 Design and construction. Shock absorbers shall conform to the manufacturer's designs and construction except for the tolerances, envelope and mounting dimensions which shall conform to Drawing 12369308 (see 4.4.2).

3.3.1 Interchangeability. All shock absorbers shall be interchangeable as to fit, form, and function with other manufacturer's assemblies having the same military part number (see 4.4.2).

3.3.2 Weight. Weight of the shock absorber shall be not greater than 50 pounds (22.7 kilograms (kg)) (see 4.4.2).

3.3.3 Positively locked piston nut. Where applicable, after torquing, the piston nut shall be secured to prevent loosening (see 4.4.2).

3.4 Protective coating. The shock absorber external surfaces shall be cleaned, treated and top coated in accordance with the manufacturer's standard practice per applicable paragraph of section 2, exterior paint system, Drawing 12350824. Painting shall be accomplished without overspray on the piston rod surfaces (see 4.4.2).

3.5 Performance requirements.

3.5.1 Damping resistance. Damping resistance shall be within the limits specified in table I (see 4.4.4.1 and 4.4.2).

TABLE I. Damping resistance.

Performance data sinusoidal motion 3-inch (7.62 cm) stroke				
Cycling rate (cpm) 1/	Resistance at midstroke			
	Compression		Rebound	
	pounds	(kg)	pounds	(kg)
47-53	3300-6800	(1497 to 3084)	2400-3300	(1089 to 1497)
94-106	5100-7100	(2313 to 3220)	2500-3600	(1134 to 1633)
141-159	5200-7300	(2359 to 3311)	2700-3900	(1225 to 1769)
282-318	5600-8200	(2540 to 3719)	3100-4700	(1406 to 2132)
564-636	6400-11,100	(2903 to 5035)	3400-9400	(1542 to 4264)

1/ Cycles per minute

NOTES: (1) Resistance at first 10 percent of each stroke shall be at least 1650 lbs (748 kilograms (kg)) in compression and 1200 lbs (544 kg) in rebound at all cycling speeds noted above.
(2) Limits apply for shock absorber temperature within ranges of 50° to 100°F (10° to 38°C) and 350° to 400°F (177° to 204°C).

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3.5.2 Compression and tension loading. The shock absorber shall be capable of withstanding a 40,000 pound (18 144 kg) compressive force and a 40,000 pound tensile force, applied through the self-aligning bearings, without affecting the damping capability (see 4.4.4.2).

3.5.3 Endurance. The shock absorber shall be capable of operating satisfactorily for 300,000 feet (91 440 meter (m)) of piston travel at a piston velocity necessary to maintain continuous fluid reservoir surface temperature (measured at lower end of reservoir) of 350° to 400°F (177° to 204°C) of piston travel at a piston velocity necessary reservoir surface temperature (measured at lower end). External cooling may be provided as required. The shock absorber shall be capable of withstanding piston velocities of 100 inches per second (2.54 meter per second (m/s)) at mid-stroke in compression and 100 inches per second at mid-stroke in rebound, obtained within 3 inches (76 mm) of the fully extended position. Total external leakage during endurance performance shall be no greater than 15 percent of the damping fluid volume prior to endurance performance. There shall be no more than 25 percent change in damping resistance after the endurance performance compared with the damping resistance prior to the endurance performance (see 4.4.4.3).

3.5.4 Bearing rotational torque. The torque required to rotate the upper, spherical inner race at 3 to 5 revolutions per minute (rpm) shall be from 12 to 100 foot-pounds (ft-lb), (16.7 to 135.6 Newton-meter (N-m). The torque required to rotate the lower spherical inner race at 3 to 5 rpm shall be from 8 to 150 ft-lb (10.8 to 203 N-m) (see 4.4.4.4 and 4.4.2).

3.5.5 Axial load. The self-aligning bearing shall not move from its installed location nor shall it be permanently deformed when an axial load of 2000 pounds (907 kg) is applied in either direction to the inner bearing shell (see 4.4.4.5).

3.5.6 Angular velocity. The self-aligning bearing shall not degrade overall performance below acceptable limits and shall retain its unit integrity when subjected to a radial load of 6000 pounds (2722 kg) and when the inner race is oscillated through a 65 $\pm$ 5 degree angle at a rate of 1 $\pm$ 0.2 cycles per second for 1 hour. One cycle equals 130 degrees of total angular motion (see 4.4.4.6).

3.6 Environmental conditions.

3.6.1 Low temperature. The shock absorber shall demonstrate no performance degradation and show no evidence of damage or deformation after being stored for not less than 8 hours at -70°F (-56.6°C). Fluid leakage shall be not greater than 1 drop per hour (see 4.4.5.1).

3.6.2 Stream and waterjet cleaning. The shock absorber shall demonstrate no performance degradation and show no evidence of damage or deformation after being subjected to steam cleaning, using high pressure steam and cleaner as specified in 4.4.5.2, followed by waterjet cleaning.

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3.7 Vehicle system testing. Vehicle system testing shall be performed to assure shock absorber compatibility with mating system, performance under vehicle operational environment and adequate reliability (see 4.5).

3.7.1 Reliability requirements. The shock absorber shall demonstrate a minimum average life of 5000 miles (8046 kilometers (km)) during vehicle testing (see 4.5.1).

3.8 Identification marking. Identification marking shall include, as a minimum, the following information permanently marked on the exterior of the shock absorber (see 4.4.2 and 6.2):

a. Manufacturer's identification
b. Serial number
c. Military part number
d. National stock number
e. Manufacturer's part number
f. Date of manufacture
g. Contract number
h. US.

3.9 Workmanship. Workmanship shall be such as to provide a shock absorber free from defects, such as: cracked, dented, loose, broken, or corroded parts or components; burrs or sharp edges; scratched finish; peeled or blistered paint (see 4.4.2).

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspections (see 4.3).

4.2 First article inspection. First article inspection shall be performed on three shock absorbers when a first article sample is required (see 3.1). First article samples shall be inspected as specified in table II (see 4.4.2). All cycling rates specified in table I shall apply to two groups of temperature ranges, Group A and B (see 4.4.4.1.2 and 4.4.4.1.3). Failure of any inspection shall be cause for rejection.

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TABLE II. Classification of inspections.

Inspection	Require -ment	Inspection Method	First article
Design and construction	3.3	SIE 1/	X
Interchangeability	3.3.1	SIE 1/	X
Weight	3.3.2	Scale	X
Positively locked system nut	3.3.3	Visual	X
Protective coating	3.4	Visual	X
Damping resistance	3.5.1	4.4.4.1	X
Compression and tension	3.5.2	4.4.4.2	X
loading	3.5.3	4.4.4.3	X
Endurance	3.5.4	4.4.4.4	X
Bearing rotational torque	3.5.5	4.4.4.5	X
Axial load	3.5.6	4.4.4.6	X
Angular velocity			
Low temperature	3.6.1	4.4.5.1	X
Steam and water jet cleaning	3.6.2	4.4.5.2	X
Vehicle System test	3.7	4.5	X
Identification marking	3.8	Visual	X
Workmanship	3.9	Visual	X

1/ SIE = Standard Inspection Equipment

4.3 Conformance inspection.

4.3.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections.

4.3.1.1 Group A inspection. Group A inspection shall consist of the inspection specified in table III.

TABLE III. Classification of inspections, Group A.

Inspection	Requirement	Method of inspection
Damping resistance	3.5.1	4.4.4.1 4.4.4.1.2

4.3.1.1.1 Group A sampling. 100 percent inspection shall apply to group A inspection.

4.3.1.2 Group B inspection. Group B inspection shall consist of the inspections specified in table IV.

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4.3.1.2.1 Group B sampling. Frequency of inspection and/or any special arrangements for establishing the frequency of inspection shall be as specified in the contract or purchase order (see 6.2). Recommended requirements for a sampling plan are shown in 6.3.

TABLE IV. Classification of defects, Group B.

Category	Inspection	Method of inspection
Major:		
101	Dimensions (see 3.3).	SIE 1/
102	Damping resistance (see 3.5.1).	4.4.4
103	Bearing rotational torque (see 3.5.4).	SIE
104	Interchangeability (see 3.3.1).	SIE
105	Weight (see 3.3.2).	Scale
Minor:		
201	Protective coating (see 3.4).	Visual
202	Identification marking (see 3.8).	Visual
203	Workmanship (see 3.9).	Visual

1/ SIE = Standard Inspection Equipment

4.4 Methods of inspection.

4.4.1 Materials. Conformance to 3.2 thru 3.2.9 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.4.1.1 Rubber components. To determine conformance to 3.2.4, rubber components, subject to ozone attack and where function and durability are impaired by the effect of ozone, shall be ozone resistant, or under stress in usage as specified in ASTM D2000.

4.4.2 Defect. Conformance to 3.3, 3.3.1, 3.3.2, 3.3.3, 3.4, 3.5.1, 3.5.4, 3.8 and 3.9 shall be determined by examination for the defect listed in table IV, examination shall be visual, tactile, or by measurement with SIE.

4.4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following standard (room) ambient conditions:

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a. Temperature: 77 ± 18°F (25° ± 10°C)
b. Relative humidity: uncontrolled room ambient
c. Atmospheric pressure: site pressure.

4.4.4 Performance.

4.4.4.1 Damping resistance. To determine conformance to 3.5.1, the shock absorber shall be mounted in a suitable test fixture by means of the self-aligning bearings and shall be cycle tested in rebound and compression. During testing, the compression and rebound damping resistance at various piston velocities shall be measured and recorded. The number of measurements taken shall be of sufficient quantity to establish that the damping resistance to the shock absorber is within the limits specified in table I. Temperatures shall be measured on the surface of the shock absorber at the lower end of the reservoir. Cycle rates and temperature ranges for first article, group A and B inspections shall be as follows:

4.4.4.1.1 First article. All cycling rates specified in table I at both temperature ranges shall apply to first article.

4.4.4.1.2 Group A. Cycling rate of 47 to 53 cpm at temperature range of 50 to 100°F (10° to 38°C) applies to Group A.

4.4.4.1.3 Group B. Cycling rates of 47 to 53 cpm and 94 to 106 cpm at both temperature ranges apply to Group B.

4.4.4.2 Compression and tension loading. To determine conformance to 3.5.2, the shock absorber shall be mounted in a suitable test fixture by means of self-aligning bearings and shall be loaded with not less than 40,000 pounds in compression and not less than 40,000 pounds in tension without damage to any of its components.

4.4.4.3 Endurance. To determine conformance to 3.5.3, the shock absorber shall be mounted in a suitable test fixture by means of the self-aligning bearings and shall be tested for not less than 600,000 cycles using a 3 ± 0.125-inch (76 ± 3 mm) stroke at a piston velocity necessary to produce a continuous fluid reservoir surface temperature of 350° to 400°F (177° to 204°C). External cooling may be provided as required. External leakage during testing shall be greater than 15 percent of the reservoir fluid when the shock absorber is fully collapsed. Following testing there shall be no more than 25 percent change in damping resistance as measured prior to the start of testing. When the damping resistance measurements are taken, the fluid reservoir surface temperature shall be 300° to 400°F (149° to 204°C).

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4.4.4.4 Bearing rotational torque. To determine conformance to 3.5.4, the upper and lower spherical inner bearing races shall be rotated at 3 to 5 rpm and the torque required to do so for each shall be measured.

4.4.4.5 Axial load. To determine conformance to 3.5.5, an axial load through the bearing of 2,000 lbs (907 kg) shall be applied to the inner bearing shell, first in one direction and then in the opposite direction. Following testing, the self-aligning bearing shall not have moved from its installed location nor shall it be permanently deformed.

4.4.4.6 Angular velocity. To determine conformance to 3.5.6, a radial load of 6,000 lbs (2722 kg) shall be applied to the self-aligning bearing. Bearing should stay in its installed location.

4.4.5 Environmental.

4.4.5.1 Low temperature. To determine conformance to 3.6.1, the shock absorber shall be tested as follows:

a. Place shock absorber in a cold chamber for not less than 8 hours at -70°F (-56.6°C).
b. Remove shock absorber from cold test chamber and mount it in a suitable test fixture by means of the self-aligning bearings. Start cycle testing within 15 minutes after removal from the cold chamber. Cycle testing shall consist of operating the shock absorber through a 3 ± 0.125-inch (76 ± 3 mm) stroke for no less than 120 cycles within 30 minutes.

   NOTE: One cycle equals 6 inches (152 mm) and 120 cycles equal 60 feet (18.3 meters (m)) of piston travel.

c. Measure and record external oil leakage during testing.

4.4.5.2 Steam and waterjet cleaning. To determine conformance to 3.6.2, the shock absorber shall be subjected to steam cleaning and a cleaner in accordance with ASTM D3399-81, followed immediately by waterjet cleaning as follows. The jet pressure shall be 105 ± 5 (723 kiloPascal (kPa) ± 34 kPa) pound-force per square inch gage (psig) for steam cleaning and 50 ± 5 (344 ± 34 kPa) psig for waterjet cleaning. Prior to and within 1 hours (unless otherwise directed by the Government) after completing the steam and waterjet cleaning, the shock absorber shall be subjected to the damping resistance test specified in 4.4.4.1. There shall be no entry of water or moisture to modules sealed against water or moisture. Moisture inside the shock absorber shall not be considered a failure.

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4.5 Vehicle systems testing. Sixteen preproduction shock absorbers shall be used for vehicle testing. Eight shock absorbers shall be installed on each of two test vehicles which shall be operated for 6000 miles (9656 km) each and shall approximate conditions of initial production tests. Tests shall include 1260 miles (2028 km) of paved roads, 2700 miles (4345 km) of secondary roads and 2040 miles (3283 km) of cross country roads. Each shock absorber shall be baseline tested in accordance with 4.4.4.1 before vehicle testing to assure conformance to the requirements of 3.5.1. After completion of 6000 miles, the damping resistance test specified in 4.4.4.1 shall be repeated and performance shall be no more than 25 percent below the minimum requirements of 3.5.1 (see 3.7).

4.5.1 Reliability requirements. The shock absorber shall meet the requirements specified in 3.7.1 during vehicle testing. The reliability calculation shall be based on the total accumulated shock absorber test miles before failure divided by the total number of shock absorbers tested. Failure is defined as shock absorber replacement required because of the inability to perform its required function. A shock absorber shall be considered a failure if one or more of the following conditions exist (see 3.7.1):

a. External leakage.
b. Little or no increase in surface temperature after cross country operation.
c. Severe bearing wear.
d. Loss of more than 25 percent of minimum damping resistance (see 3.5.1).

If a shock absorber has operated for 6000 miles without replacement, its life shall be taken as 6000 miles for the reliability calculation.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. Shock absorbers manufactured in accordance with this specification are intended for use on military tracked vehicles in the BFV family.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2 and 2.3).
c. When first article is required (see 3.1).
d. Identification and marking if other than as specified (see 3.8).
e. Sampling plan requirements for Group B inspection (see 4.3.1.2.1).
f. Inspection conditions if other than as specified (see 4.4.3).
g. Packaging requirements (see 5.1).

6.3 Recommended group B sampling plan (see 4.3.1.2.1).

6.3.1 Basic method. Quality characteristics listed in table III will be inspected using the sampling plan shown in table V. Alternate sampling plans are not allowed except under an approved statistical process control (SPC) system conforming to the requirements outlined in paragraph 6.4.2.

TABLE V. Group B sampling plan.

Lot size	Sample size	
	Major characteristics	Minor characteristics
2 to 13	Entire lot	3
14 to 25	13	3
26 to 50	13	5
51 to 90	13	6
91 to 150	13	8
151 to 280	20	10
281 to 500	29	11
501 to 1200	34	15
1201 to 3200	42	19
3201 to 10,000	50	22

NOTE: Accept lot on zero defects.
Reject lot on one or more defects.

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6.3.1.1 Tightened inspection. Tightened inspection should be introduced as soon as 2 out of 5 successive lots have been rejected and should as a minimum, impose a 30 percent increase in sample size. Normal inspection sampling may be restored after 5 successive lots have been accepted under tightened inspection.

6.3.1.2 Reduced inspection. Reduced inspection may be introduced when 10 successive lots have been accepted and should as a maximum, permit a 30 percent decrease in sample size. Normal inspection sampling should be restored if a lot is not accepted under reduced inspection.

6.3.1.3 Statistical process control (SPC). Sampling plan requirements may be satisfied through the use of SPC as an alternate control method. If SPC is to be used, an SPC plan, detailing the methods for monitoring process control, should be approved by the procuring activity or agent. The plan, developed by the contractor, should include as a minimum:

a. Types of control charts and their uses.
b. Process capability (C_{pk}) studies.
c. Criteria for determining out-of-control conditions.
d. Corrective actions to be taken if an out-of-control or out-of-tolerance condition is detected.
e. Training programs and qualification requirements of personnel executing the plan.
f. Provisions for reduced inspection.

The plan should contain the results of process capability studies for the processes being controlled. For variable data, the minimum C_{pk} for statistical control should be 1.33. For attribute data, a minimum process average of 99.73% is required.

6.4 Definitions.

6.4.1 Direct acting shock absorber. The shock absorber is a self-contained, double-acting, energy-absorbing, hydraulic unit. Included as part of the shock absorber are three concentric cylinders of which the innermost cylinder contains a double-acting piston that divides the cylinder into upper and lower chambers. The annular volume between the innermost cylinder and the middle cylinder serves as a fluid reservoir. The outermost cylinder serves as a shield to protect the piston rod from damage. The top of the shield has ventilation holes and is covered with a flexible rubber baffle to limit debris entrance through the holes and to permit the exit of mud. Compression and rebound valves are contained in the piston which is fastened to the actuation rod. One end of the innermost cylinder is formed by the base valve, which contains a compressive and an intake valve. Mounting of the shock absorber is provided by two self-aligning bearings, one at each end.

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6.5 Subject term (key word) listing.

Axial load
Bearing
Compression
Cycling rate
Damping resistance
Torque
Valve spring.

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army - AT

Preparing Activity:
Army - AT

(Project 2540-0421)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-62494C(AT)	2. DOCUMENT DATE (YYMMDD) 951026

3. DOCUMENT TITLE
SHOCK ABSORBER, DIRECT ACTING

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (810) 574-8745	(2) AUTOVON 786-8745
c. ADDRESS (Include Zip Code) Commander U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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